

Solutions of Problems on ch 23

4. A charged particle A exerts a force of $2.62 \mu\text{N}$ to the right on charged particle B when the particles are 13.7 mm apart. Particle B moves straight away from A to make the distance between them 17.7 mm . What vector force does it then exert on A ?

$$\vec{\mathbf{F}}_{B \text{ on } A,2} = -\vec{\mathbf{F}}_{A \text{ on } B} = \frac{k_e |q_A| |q_B|}{r_2^2} (-\hat{\mathbf{i}})$$

$$\frac{F_2}{F_1} = \frac{k_e |q_A| |q_B|}{r_2^2} \frac{r_1^2}{k_e |q_A| |q_B|}$$

$$F_2 = \frac{F_1 r_1^2}{r_2^2} = (2.62 \mu\text{N}) \left(\frac{13.7 \text{ mm}}{17.7 \text{ mm}} \right)^2 = 1.57 \mu\text{N}$$

$$\vec{\mathbf{F}}_{B \text{ on } A,2} = \boxed{1.57 \mu\text{N to the left}}.$$

- 11.** Three point charges are arranged as shown in Figure P23.11. Find (a) the magnitude and (b) the direction of the electric force on the particle at the origin.

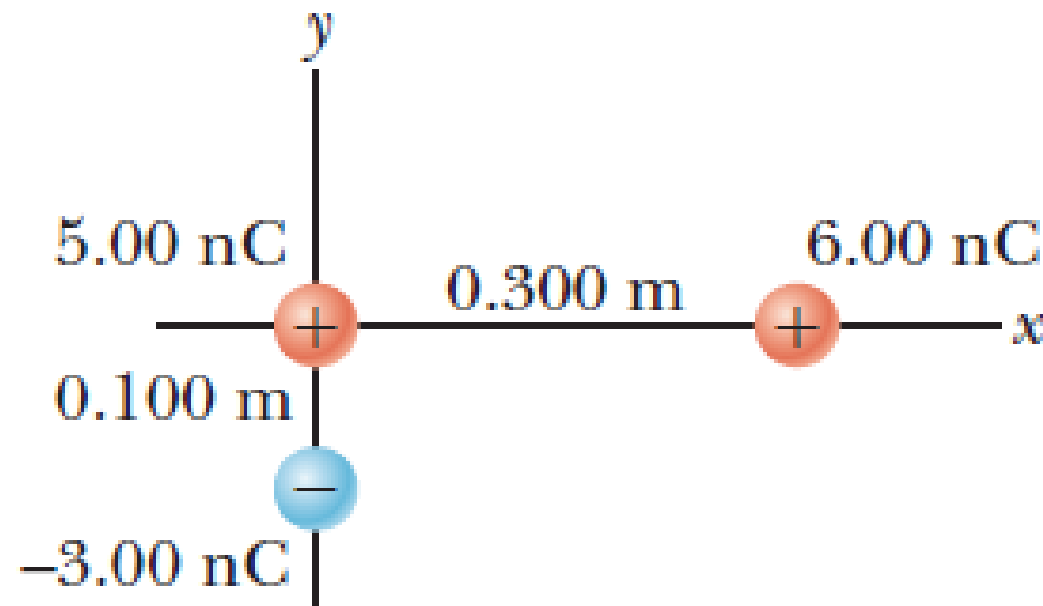
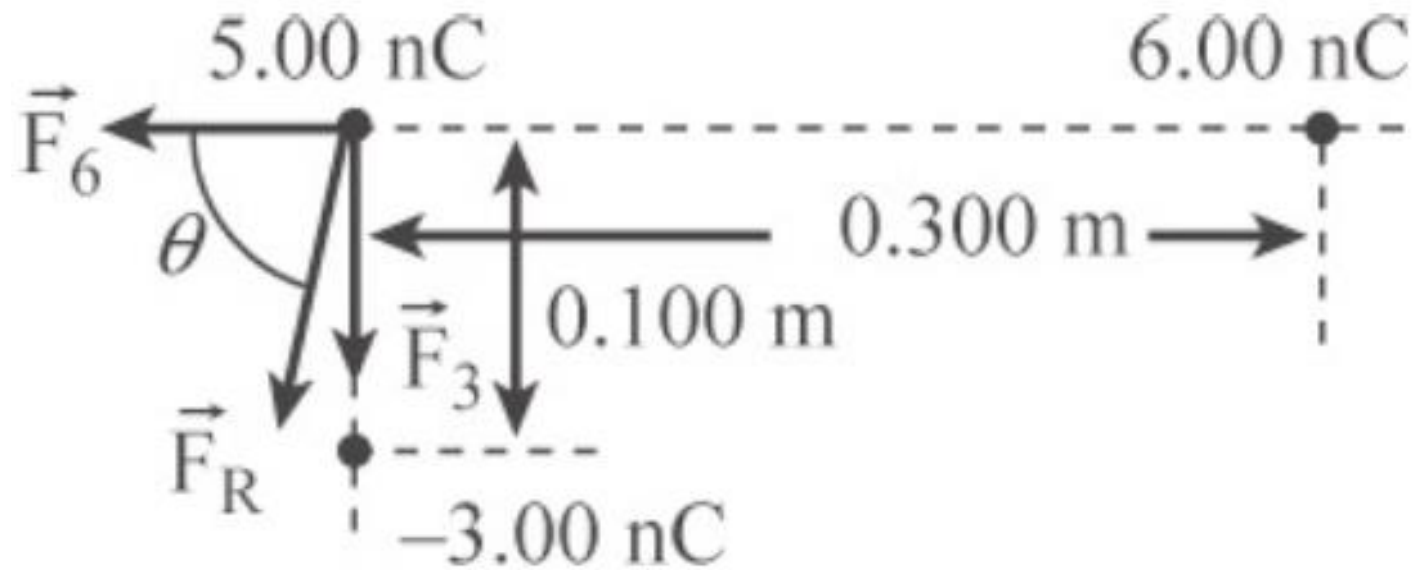


Figure P23.11 Problems 11 and 35.



$$F_x = -F_6 = -\left(8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}\right) \frac{(6.00 \times 10^{-9} \text{ C})(5.00 \times 10^{-9} \text{ C})}{(0.300 \text{ m})^2}$$

$$= -3.00 \times 10^{-6} \text{ N} \quad (\text{to the left})$$

$$F_y = -F_3 = -\left(8.99 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}\right) \frac{(3.00 \times 10^{-9} \text{ C})(5.00 \times 10^{-9} \text{ C})}{(0.100 \text{ m})^2}$$

$$= -1.35 \times 10^{-5} \text{ N} \quad (\text{downward})$$

- (a) The forces are perpendicular, so the magnitude of the resultant is

$$F_R = \sqrt{(F_6)^2 + (F_3)^2} = \boxed{1.38 \times 10^{-5} \text{ N}}$$

- (b) The magnitude of the angle of the resultant is

$$\theta = \tan^{-1} \left(\frac{F_3}{F_6} \right) = 77.5^\circ$$

The resultant force is in the third quadrant, so the direction is

$$\boxed{77.5^\circ \text{ below } -x \text{ axis}}$$

- 13.** Two small beads having positive charges $q_1 = 3q$ and $q_2 = q$ are fixed at the opposite ends of a horizontal insulating rod of length $d = 1.50$ m. The bead with charge q_1 is at the origin. As shown in Figure P23.13, a third small, charged bead is free to slide on the rod.
- W** (a) At what position x is the third bead in equilibrium?
- (b) Can the equilibrium be stable?

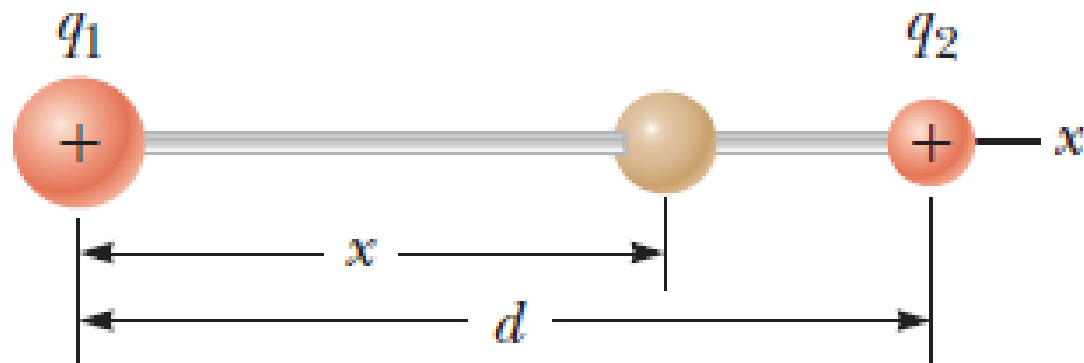


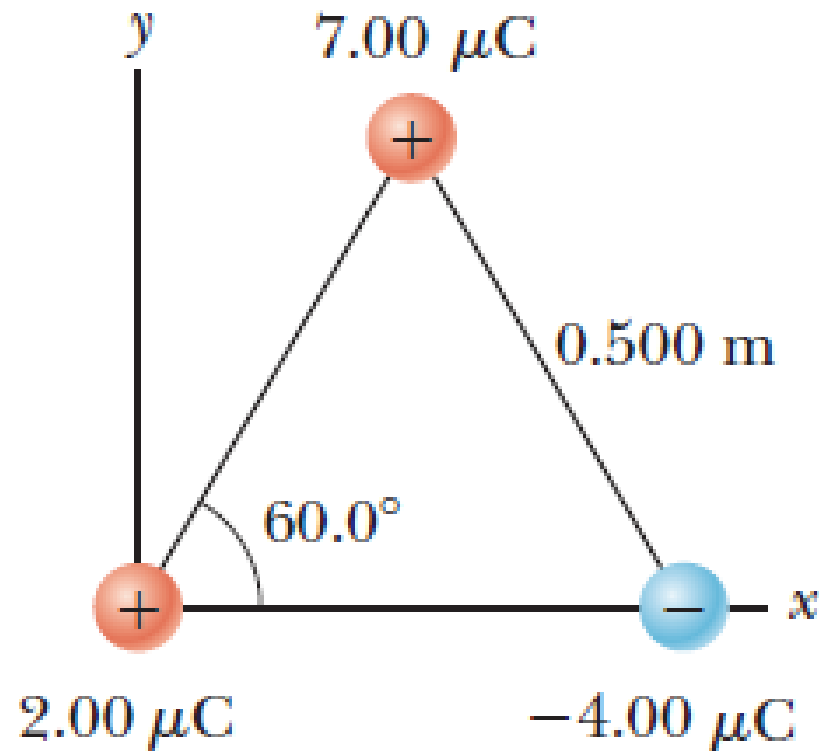
Figure P23.13 Problems 13 and 14.

$$\vec{F} = \frac{k_e(3q)Q}{x^2}\hat{\mathbf{i}} + \frac{k_e(q)Q}{(d-x)^2}(-\hat{\mathbf{i}}), \text{ where } d = 1.50 \text{ m}$$

$$\frac{3}{x^2} = \frac{1}{(d-x)^2}, \text{ or } d-x = \frac{x}{\sqrt{3}}$$

$$x = 0.634d = 0.634(1.50 \text{ m}) = \boxed{0.951 \text{ m}}$$

- 15.** Three charged particles are located at the corners of an equilateral triangle as shown in Figure P23.15. Calculate the total electric force on the $7.00\text{-}\mu\text{C}$ charge.



$$\vec{F}_1 = k_e \frac{q_1 q_2}{r^2} \hat{r} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(7.00 \times 10^{-6} \text{ C})(2.00 \times 10^{-6} \text{ C})}{(0.500 \text{ m})^2} \times (\cos 60^\circ \hat{i} + \sin 60^\circ \hat{j})$$

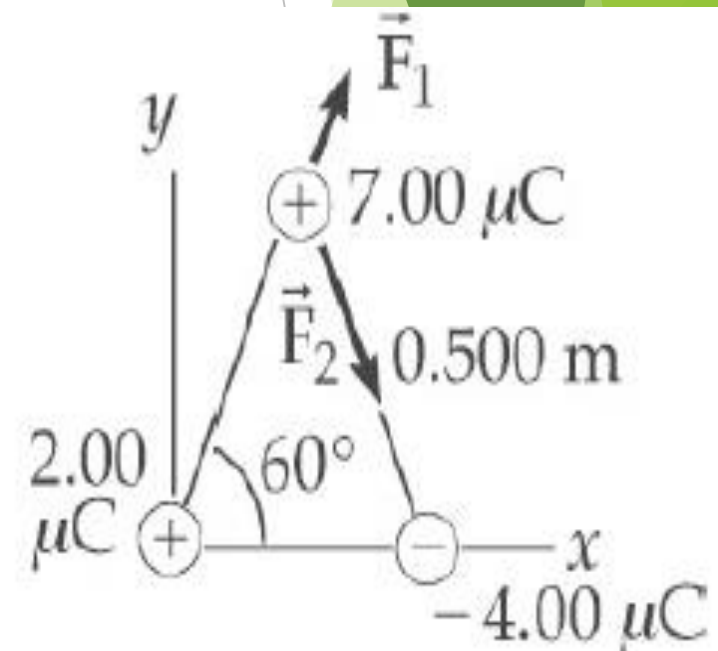
$$\vec{F}_1 = (0.252 \hat{i} + 0.436 \hat{j}) \text{ N}$$

$$\vec{F}_2 = k_e \frac{q_1 q_3}{r^2} \hat{r} = -\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(7.00 \times 10^{-6} \text{ C})(-4.00 \times 10^{-6} \text{ C})}{(0.500 \text{ m})^2} \times (\cos 60^\circ \hat{i} - \sin 60^\circ \hat{j})$$

$$\vec{F}_2 = (0.503 \hat{i} - 0.872 \hat{j}) \text{ N}$$

$$\vec{F} = \vec{F}_1 + \vec{F}_2 = (0.755 \hat{i} - 0.436 \hat{j}) \text{ N}$$

$$\vec{F} = (0.755 \text{ N}) \hat{i} - (0.436 \text{ N}) \hat{j} = \boxed{0.872 \text{ N at an angle of } 330^\circ}$$



21. Two identical conducting small spheres are placed with their centers 0.300 m apart. One is given a charge of 12.0 nC and the other a charge of -18.0 nC. (a) Find the electric force exerted by one sphere on the other. (b) **What If?** The spheres are connected by a conducting wire. Find the electric force each exerts on the other after they have come to equilibrium.

$$F = \frac{k_e q_1 q_2}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(12.0 \times 10^{-9} \text{ C})(18.0 \times 10^{-9} \text{ C})}{(0.300 \text{ m})^2} = \boxed{2.16 \times 10^{-5} \text{ N}}$$

$$F = \frac{k_e q_1 q_2}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(3.00 \times 10^{-9} \text{ C})(3.00 \times 10^{-9} \text{ C})}{(0.300 \text{ m})^2} = \boxed{8.99 \times 10^{-7} \text{ N}}$$

24. A small object of mass 3.80 g and charge $-18.0 \mu\text{C}$ is suspended motionless above the ground when immersed in a uniform electric field perpendicular to the ground. What are the magnitude and direction of the electric field?

$$E = \frac{mg}{|q|} = \frac{(3.80 \times 10^{-3} \text{ kg})(9.80 \text{ m/s}^2)}{18.0 \times 10^{-6} \text{ C}} = \boxed{2.07 \times 10^3 \text{ N/C}}$$

25. Four charged particles are at the corners of a square of side a as shown in Figure P23.25. Determine (a) the electric field at the location of charge q and (b) the total electric force exerted on q .

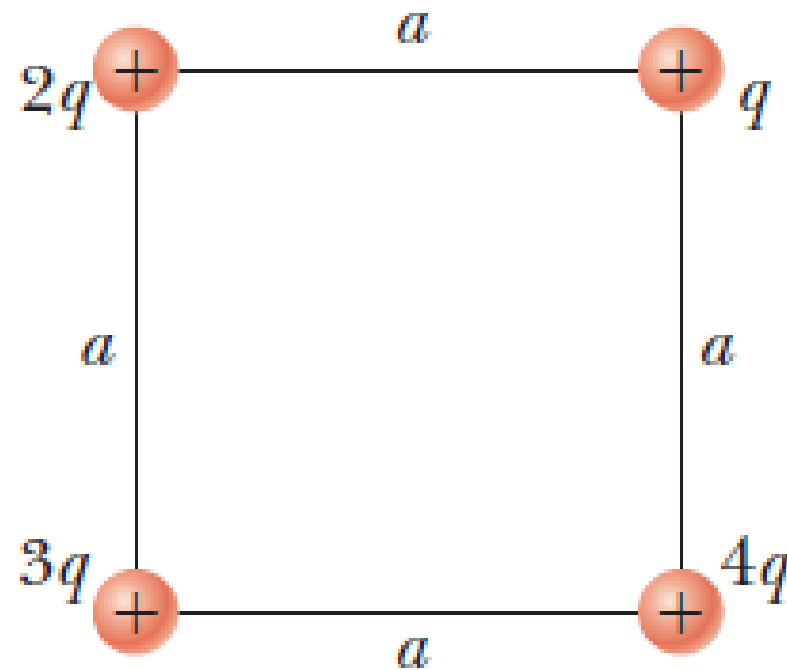


Figure P23.25

$$\vec{\mathbf{E}} = \frac{k_e q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{k_e q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{k_e q_3}{r_3^2} \hat{\mathbf{r}}_3$$

(a) Substituting for each of the charges gives

$$\begin{aligned}\vec{\mathbf{E}} &= \frac{k_e (2q)}{a^2} \hat{\mathbf{i}} + \frac{k_e (3q)}{2a^2} (\hat{\mathbf{i}} \cos 45.0^\circ + \hat{\mathbf{j}} \sin 45.0^\circ) + \frac{k_e (4q)}{a^2} \hat{\mathbf{j}} \\ &= \frac{k_e q}{a^2} \left[\left(2 + \frac{3}{2} \cos 45.0^\circ \right) \hat{\mathbf{i}} + \left(\frac{3}{2} \sin 45.0^\circ + 4 \right) \hat{\mathbf{j}} \right] \\ &= \boxed{\frac{k_e q}{a^2} (3.06 \hat{\mathbf{i}} + 5.06 \hat{\mathbf{j}})}\end{aligned}$$

(b) The electric force on charge q is given by

$$\vec{\mathbf{F}} = q \vec{\mathbf{E}} = \boxed{\frac{k_e q^2}{a^2} (3.06 \hat{\mathbf{i}} + 5.06 \hat{\mathbf{j}})}$$

26. Three point charges lie along a circle of radius r at angles of 30° , 150° , and 270° as shown in Figure P23.26. Find a symbolic expression for the resultant electric field at the center of the circle.

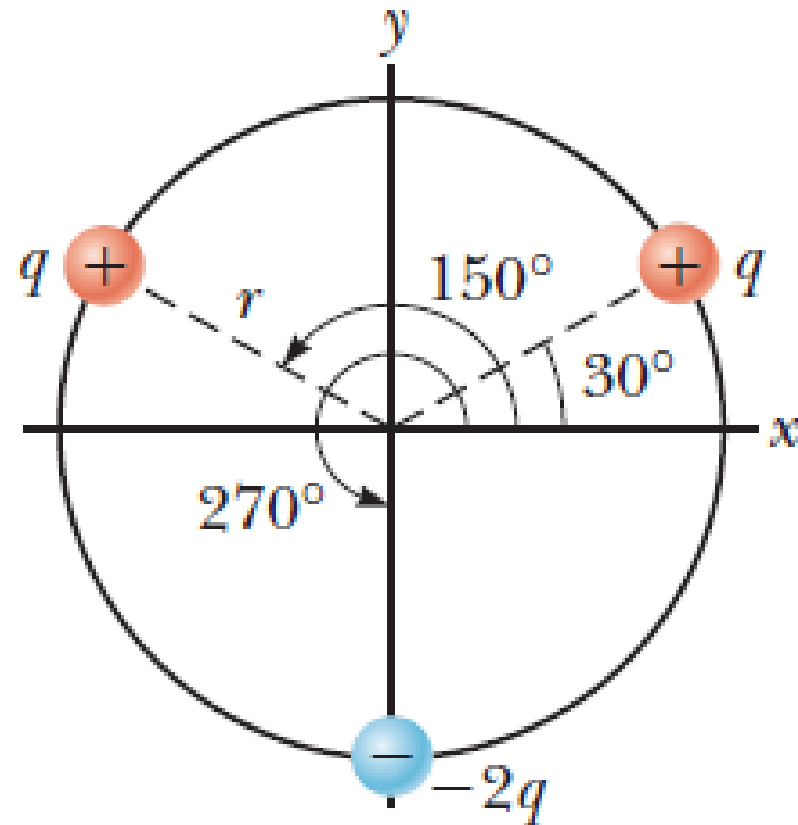
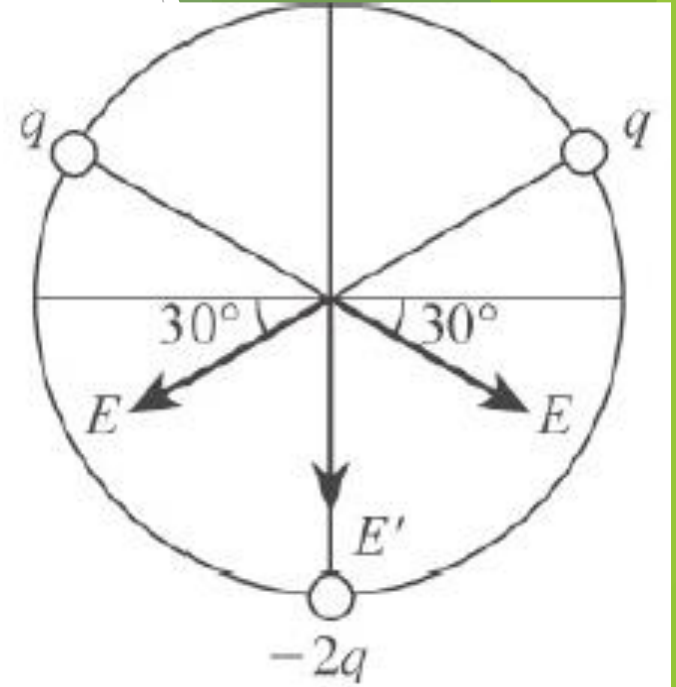


Figure P23.26

$$E = \frac{k_e q}{r^2} \text{ and } E' = \frac{k_e (2q)}{r^2} = 2E$$

$$\begin{aligned}\vec{E} &= (E \cos 30.0^\circ - E \cos 30.0^\circ) \hat{i} - (E' + 2E \sin 30.0^\circ) \hat{j} \\ &= -(E' + 2E \sin 30.0^\circ) \hat{j} = -(2E + 2E \sin 30.0^\circ) \hat{j} \\ &= -2E(1 + \sin 30.0^\circ) \hat{j} \\ &= -2 \frac{k_e q}{r^2} (1 + \sin 30.0^\circ) \hat{j} = -2 \frac{k_e q}{r^2} (1.50) \hat{j} = \boxed{-k_e \frac{3q}{r^2} \hat{j}}\end{aligned}$$





29. In Figure P23.29, determine the point (other than infinity) at which the electric field is zero.

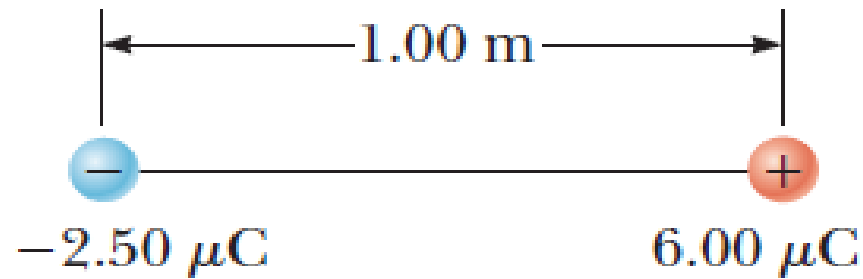


Figure P23.29

$$\frac{k_e |q_-|}{x^2} = \frac{k_e |q_+|}{(1 \text{ m} + x)^2} \quad \Longrightarrow \quad |q_-|^{1/2} (1 \text{ m} + x) = q_+^{1/2} x$$

$$1 \text{ m} + x = \left(\frac{6.00}{2.50} \right)^{1/2} x = 1.55 x \quad \Longrightarrow \quad 1 \text{ m} = 0.549 x$$

$$x = \boxed{1.82 \text{ m}}$$

- 31.** Three point charges are located on a circular arc as shown in Figure P23.31. (a) What is the total electric field at P , the center of the arc? (b) Find the electric force that would be exerted on a -5.00-nC point charge placed at P .

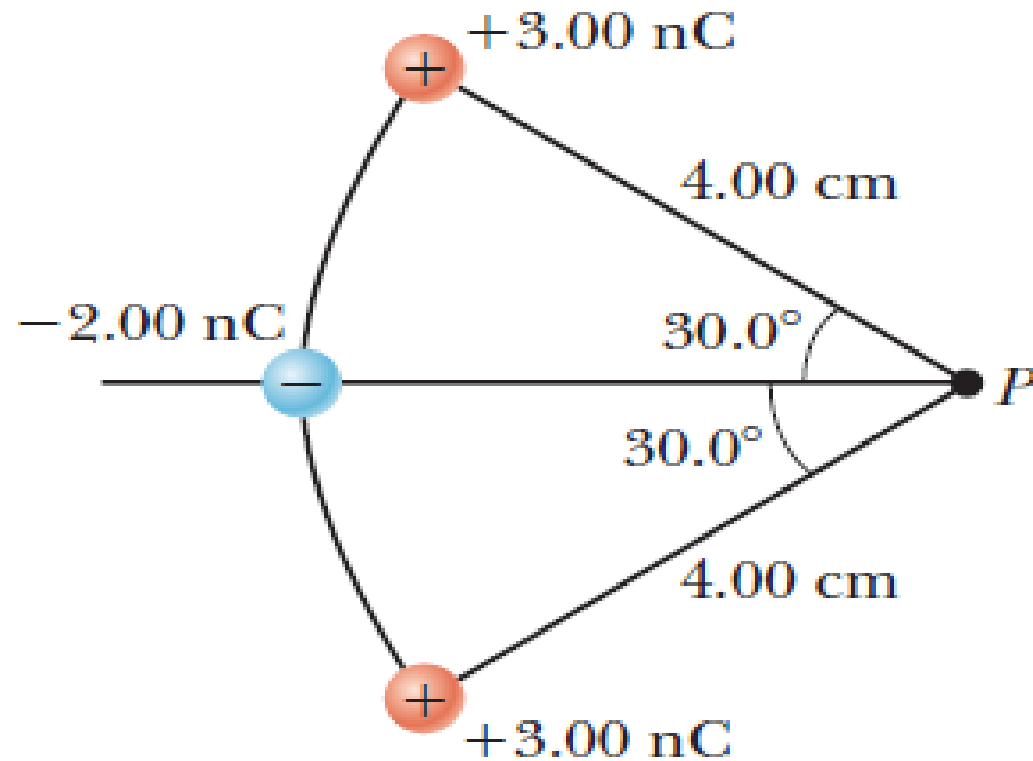


Figure P23.31

Call $Q = 3.00 \text{ nC}$ and $q = |-2.00 \text{ nC}| = 2.00 \text{ nC}$,
and $r = 4.00 \text{ cm} = 0.0400 \text{ m}$. Then,

$$E_1 = E_2 = \frac{k_e Q}{r^2} \quad \text{and} \quad E_3 = \frac{k_e q}{r^2}$$

Then,

$$E_y = 0$$

$$E_x = E_{\text{total}} = 2 \frac{k_e Q}{r^2} \cos 30.0^\circ - \frac{k_e q}{r^2}$$

$$E_x = \frac{k_e}{r^2} (2Q \cos 30.0^\circ - q)$$

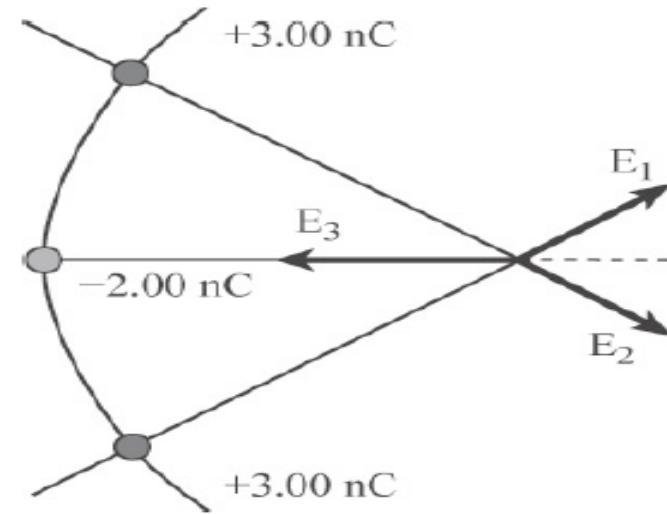
$$E_x = \left[\frac{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2}{(0.0400 \text{ m})^2} \right] \times [2(3.00 \times 10^{-9} \text{ C}) \cos 30.0^\circ - 2.00 \times 10^{-9} \text{ C}]$$

$$= 1.80 \times 10^4 \text{ N/C}$$

(a) $\boxed{1.80 \times 10^4 \text{ N/C to the right}}$

(b) The electric force on a point charge placed at point P is

$$F = qE = (-5.00 \times 10^{-9} \text{ C})E = \boxed{-8.98 \times 10^{-5} \text{ N (to the left)}}$$



ANS. FIG. P23.31

- 33.** A small, 2.00-g plastic ball is suspended by a 20.0-cm-long string in a uniform electric field as shown in Figure P23.33. If the ball is in equilibrium when the string makes a 15.0° angle with the vertical, what is the net charge on the ball?

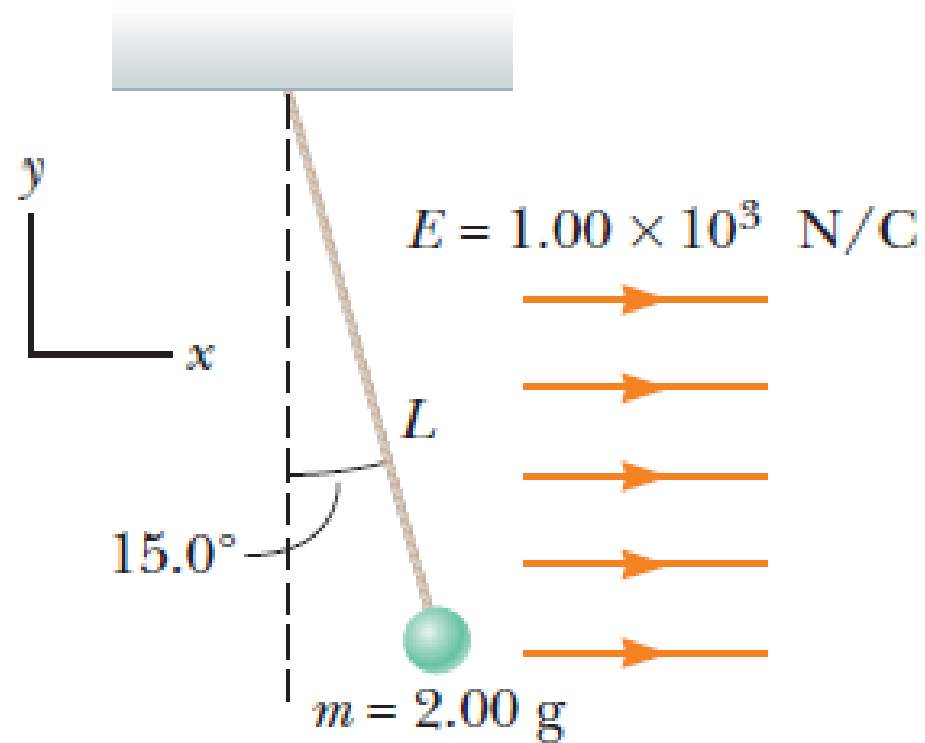


Figure P23.33

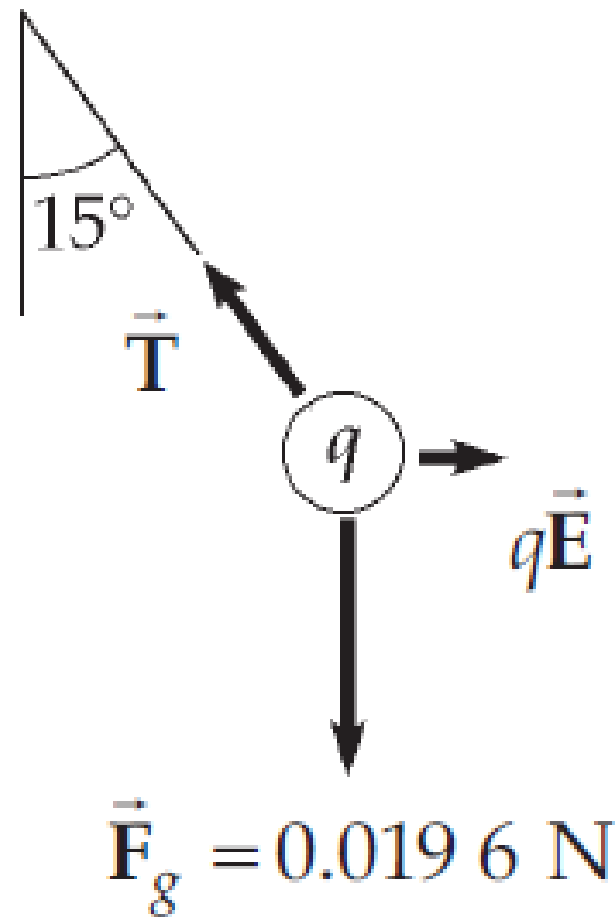
$$\sum F_y = 0: \quad T \cos 15.0^\circ = 1.96 \times 10^{-2} \text{ N}$$

$$T = 2.03 \times 10^{-2} \text{ N}.$$

$$\sum F_x = 0, \text{ we have } qE = T \sin 15.0^\circ$$

$$q = \frac{T \sin 15.0^\circ}{E} = \frac{(2.03 \times 10^{-2} \text{ N}) \sin 15.0^\circ}{1.00 \times 10^3 \text{ N/C}}$$

$$= 5.25 \times 10^{-6} \text{ C} = \boxed{5.25 \mu\text{C}}$$



- 34.** Two $2.00\text{-}\mu\text{C}$ point charges are located on the x axis. One is at $x = 1.00\text{ m}$, and the other is at $x = -1.00\text{ m}$. (a) Determine the electric field on the y axis at $y = 0.500\text{ m}$. (b) Calculate the electric force on a $-3.00\text{-}\mu\text{C}$ charge placed on the y axis at $y = 0.500\text{ m}$.

$$d = \sqrt{(1.00\text{ m})^2 + (0.500\text{ m})^2} = 1.12\text{ m}$$

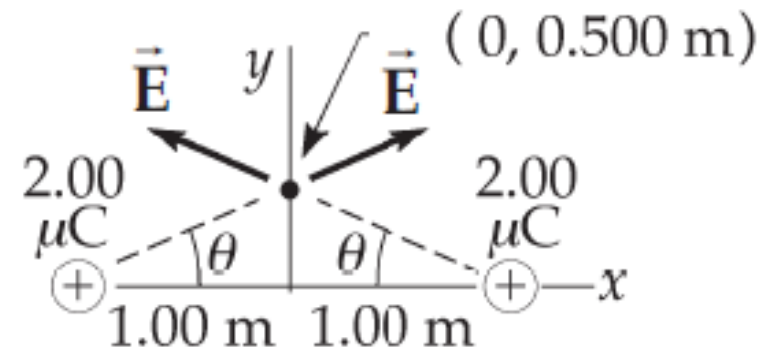
$$E = \frac{k_e q}{r^2} = \frac{(8.99 \times 10^9\text{ N}\cdot\text{m}^2/\text{C}^2)(2.00 \times 10^{-6}\text{ C})}{(1.12\text{ m})^2} = 14\,400\text{ N/C}$$

$$E_x = 0 \text{ and } E_y = 2(14\,400\text{ N/C})\sin 26.6^\circ = 1.29 \times 10^4\text{ N/C}$$

$$\boxed{\vec{E} = 1.29 \times 10^4 \hat{j}\text{ N/C}}$$

$$\vec{F} = q\vec{E} = (-3.00 \times 10^{-6}\text{ C})(1.29 \times 10^4\text{ N/C}\hat{j})$$

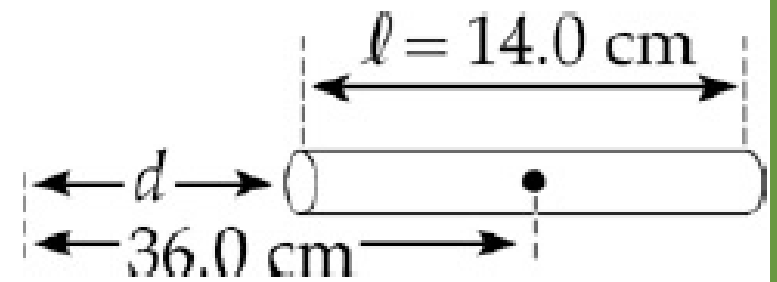
$$= \boxed{-3.86 \times 10^{-2} \hat{j}\text{ N}}$$



37. A rod 14.0 cm long is uniformly charged and has a total charge of $-22.0 \mu\text{C}$. Determine (a) the magnitude and (b) the direction of the electric field along the axis of the rod at a point 36.0 cm from its center.

$$\begin{aligned} |E| &= \frac{k_e \lambda \ell}{d(\ell + d)} = \frac{k_e (Q / \ell) \ell}{d(\ell + d)} = \frac{k_e Q}{d(\ell + d)} \\ &= \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(22.0 \times 10^{-6} \text{ C})}{(0.290 \text{ m})(0.140 \text{ m} + 0.290 \text{ m})} \end{aligned}$$

$$E = \boxed{1.59 \times 10^6 \text{ N/C}}$$



38. A uniformly charged disk of radius 35.0 cm carries charge with a density of $7.90 \times 10^{-3} \text{ C/m}^2$. Calculate the electric field on the axis of the disk at (a) 5.00 cm, (b) 10.0 cm, (c) 50.0 cm, and (d) 200 cm from the center of the disk.

$$E = 2\pi k_e \sigma \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$$

$$\begin{aligned} E &= 2\pi (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) (7.90 \times 10^{-3} \text{ C/m}^2) \times \left(1 - \frac{x}{\sqrt{x^2 + (0.350)^2}} \right) \\ &= (4.46 \times 10^8 \text{ N/C}) \left(1 - \frac{x}{\sqrt{x^2 + 0.123}} \right) \end{aligned}$$

(a) At $x = 0.0500 \text{ m}$,

$$E = 3.83 \times 10^8 \text{ N/C} = \boxed{383 \text{ MN/C}}$$

(b) At $x = 0.100 \text{ m}$,

$$E = 3.24 \times 10^8 \text{ N/C} = \boxed{324 \text{ MN/C}}$$

(c) At $x = 0.500 \text{ m}$,

$$E = 8.07 \times 10^7 \text{ N/C} = \boxed{80.7 \text{ MN/C}}$$

(d) At $x = 2.000 \text{ m}$,

$$E = 6.68 \times 10^8 \text{ N/C} = \boxed{6.68 \text{ MN/C}}$$

- 39.** A uniformly charged ring of radius 10.0 cm has a total charge of $75.0 \mu\text{C}$. Find the electric field on the axis of the ring at (a) 1.00 cm, (b) 5.00 cm, (c) 30.0 cm, and (d) 100 cm from the center of the ring.

$$\begin{aligned} |E| &= \frac{k_e x Q}{(x^2 + a^2)^{3/2}} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(75.0 \times 10^{-6} \text{ C})(0.100 \text{ m})}{(x^2 + 0.100^2)^{3/2}} \\ &= \frac{6.74 \times 10^5 x}{(x^2 + 0.0100)^{3/2}} \end{aligned}$$

(a) At $x = 0.010\ 0\ \text{m}$, $\vec{E} = 6.64 \times 10^6 \hat{i}\ \text{N/C} = \boxed{6.64 \hat{i}\ \text{MN/C}}$

(b) At $x = 0.050\ 0\ \text{m}$, $\vec{E} = 2.41 \times 10^7 \hat{i}\ \text{N/C} = \boxed{24.1 \hat{i}\ \text{MN/C}}$

(c) At $x = 0.300\ \text{m}$, $\vec{E} = 6.40 \times 10^6 \hat{i}\ \text{N/C} = \boxed{6.40 \hat{i}\ \text{MN/C}}$

(d) At $x = 1.00\ \text{m}$, $\vec{E} = 6.64 \times 10^5 \hat{i}\ \text{N/C} = \boxed{0.664 \hat{i}\ \text{MN/C}}$

51. A proton accelerates from rest in a uniform electric field of 640 N/C. At one later moment, its speed is 1.20 Mm/s (nonrelativistic because v is much less than the speed of light). (a) Find the acceleration of the proton. (b) Over what time interval does the proton reach this speed? (c) How far does it move in this time interval? (d) What is its kinetic energy at the end of this interval?

AMT

M

$$(a) \quad a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.602 \times 10^{-19} \text{ C})(640 \text{ N/C})}{1.67 \times 10^{-27} \text{ kg}} = \boxed{6.14 \times 10^{10} \text{ m/s}^2}$$

$$(b) \quad v_f = v_i + at$$

$$t = \frac{v_f - 0}{a} = \frac{1.20 \times 10^6 \text{ m/s}}{6.14 \times 10^{10} \text{ m/s}^2} = \boxed{19.5 \mu\text{s}}$$

$$(c) \quad \Delta x = v_i t + \frac{1}{2} a t^2 = 0 + \frac{1}{2} (6.14 \times 10^{10} \text{ m/s}^2) (19.5 \times 10^{-6} \text{ s})^2 \\ = \boxed{11.7 \text{ m}}$$

$$(d) \quad K = \frac{1}{2} m v^2 = \frac{1}{2} (1.67 \times 10^{-27} \text{ kg}) (1.20 \times 10^6 \text{ m/s})^2 = \boxed{1.20 \times 10^{-15} \text{ J}}$$

52. A proton is projected in the positive x direction **W** into a region of a uniform electric field $\vec{E} = (-6.00 \times 10^5) \hat{i} \text{ N/C}$ at $t = 0$. The proton travels 7.00 cm as it comes to rest. Determine (a) the acceleration of the proton, (b) its initial speed, and (c) the time interval over which the proton comes to rest.

$$(a) \quad |a| = \frac{qE}{m} = \frac{(1.602 \times 10^{-19} \text{ C})(6.00 \times 10^5 \text{ N/C})}{1.67 \times 10^{-27} \text{ kg}} = 5.76 \times 10^{13} \text{ m/s}^2,$$

$$\text{so } \vec{a} = \boxed{-5.76 \times 10^{13} \hat{i} \text{ m/s}^2}.$$

$$(b) \quad v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$0 = v_i^2 + 2(-5.76 \times 10^{13} \text{ m/s}^2)(0.0700 \text{ m})$$

$$\vec{v}_i = \boxed{2.84 \times 10^6 \hat{i} \text{ m/s}}$$

$$(c) \quad v_f = v_i + at$$

$$0 = 2.84 \times 10^6 \text{ m/s} + (-5.76 \times 10^{13} \text{ m/s}^2)t \rightarrow t = \boxed{4.93 \times 10^{-8} \text{ s}}$$

56. Two horizontal metal plates, each 10.0 cm square, are aligned 1.00 cm apart with one above the other. They are given equal-magnitude charges of opposite sign so that a uniform downward electric field of $2.00 \times 10^3 \text{ N/C}$ exists in the region between them. A particle of mass $2.00 \times 10^{-16} \text{ kg}$ and with a positive charge of $1.00 \times 10^{-6} \text{ C}$ leaves the center of the bottom negative plate with an initial speed of $1.00 \times 10^5 \text{ m/s}$ at an angle of 37.0° above the horizontal. (a) Describe the trajectory of the particle. (b) Which plate does it strike? (c) Where does it strike, relative to its starting point?

$$(a) \quad \vec{F} = q\vec{E} = (1.00 \times 10^{-6} \text{ C})(2000 \text{ N/C})(-\hat{j}) = 2.00 \times 10^{-3}(-\hat{j}) \text{ N}$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{(2.00 \times 10^{-3} \text{ N})(-\hat{j})}{2.00 \times 10^{-16} \text{ kg}} = 1.00 \times 10^{13}(-\hat{j}) \text{ m/s}^2$$

$$(b) \quad v_{yf}^2 = v_{yi}^2 + 2a_y(y_f - y_i)$$

$$y_f - y_i = \frac{v_{yf}^2 - v_{yi}^2}{2a_y} = \frac{0 - [(1.00 \times 10^5 \text{ m/s}) \sin 37.0^\circ]^2}{2(1.00 \times 10^{13} \text{ m/s}^2)} \\ = 1.81 \times 10^{-4} \text{ m} = 0.181 \text{ mm}$$

$$(c) \quad (1.00 \times 10^5 \text{ m/s}) \cos 37^\circ = 7.99 \times 10^4 \text{ m/s}$$

$$y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

$$0 = 0 + [(1.00 \times 10^5 \text{ m/s}) \sin 37.0^\circ]t + \frac{1}{2}(-1.00 \times 10^{13} \text{ m/s}^2)t^2$$

$$t = 1.20 \times 10^{-8} \text{ s.}$$

$$x_f = x_i + v_x t = 0 + (7.99 \times 10^4 \text{ m/s})(1.20 \times 10^{-8} \text{ s}) \\ = 9.61 \times 10^{-4} \text{ m}$$

57. A proton moves at 4.50×10^5 m/s in the horizontal direction. It enters a uniform vertical electric field with a magnitude of 9.60×10^3 N/C. Ignoring any gravitational effects, find (a) the time interval required for the proton to travel 5.00 cm horizontally, (b) its vertical displacement during the time interval in which it travels 5.00 cm horizontally, and (c) the horizontal and vertical components of its velocity after it has traveled 5.00 cm horizontally.

$\vec{E}$ is directed along the y direction; therefore, $a_x = 0$ and $x = v_{xi}t$.

$$(a) \quad t = \frac{x}{v_{xi}} = \frac{0.0500 \text{ m}}{4.50 \times 10^5 \text{ s}} = 1.11 \times 10^{-7} \text{ s} = \boxed{111 \text{ ns}}$$

$$(b) \quad a_y = \frac{qE}{m} = \frac{(1.602 \times 10^{-19} \text{ C})(9.60 \times 10^3 \text{ N/C})}{1.67 \times 10^{-27} \text{ kg}} = 9.21 \times 10^{11} \text{ m/s}^2$$

$$y_f - y_i = v_{yi}t + \frac{1}{2}a_y t^2:$$

$$\begin{aligned} y_f &= \frac{1}{2}(9.21 \times 10^{11} \text{ m/s}^2)(1.11 \times 10^{-7} \text{ s})^2 \\ &= 5.68 \times 10^{-3} \text{ m} = \boxed{5.67 \text{ mm}} \end{aligned}$$

$$(c) \quad v_x = 4.50 \times 10^5 \text{ m/s}$$

$$v_{yf} = v_{yi} + a_y t = (9.21 \times 10^{11} \text{ m/s}^2)(1.11 \times 10^{-7} \text{ s}) = 1.02 \times 10^5 \text{ m/s}$$

$$\vec{v} = \boxed{(450\hat{i} + 102\hat{j}) \text{ km/s}}$$